

Earth and Space Science Mid-Unit Task 1



Earth and Space Science Unit 1 Mid-Unit Formative Task A Star's Life

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Student Task

A Star's Life

Base your answers to questions 1 through 6 on the information below and on your knowledge of earth and space science.

The sun is the only star in our solar system, but in our galaxy, it is one out of 100 billion. Some of these stars are bigger than our sun, and some are smaller than our sun.

Image 1: Our Sun's Nearest Neighbors

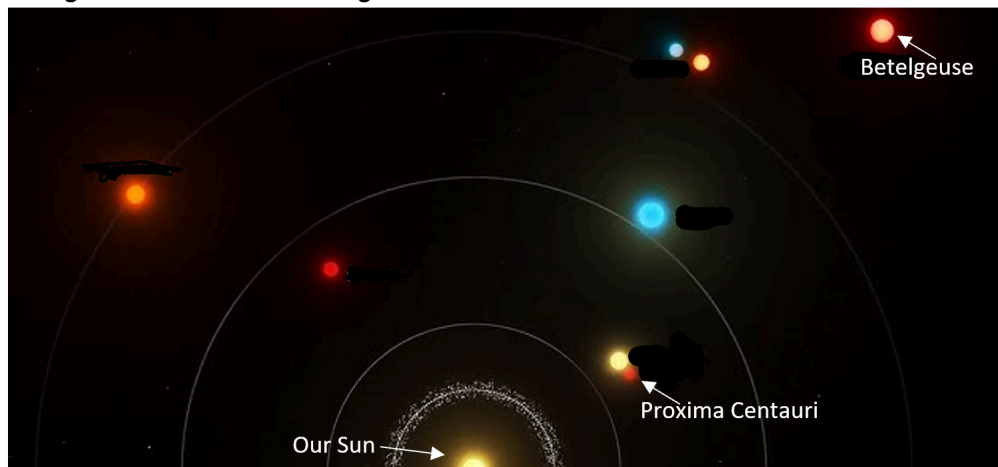


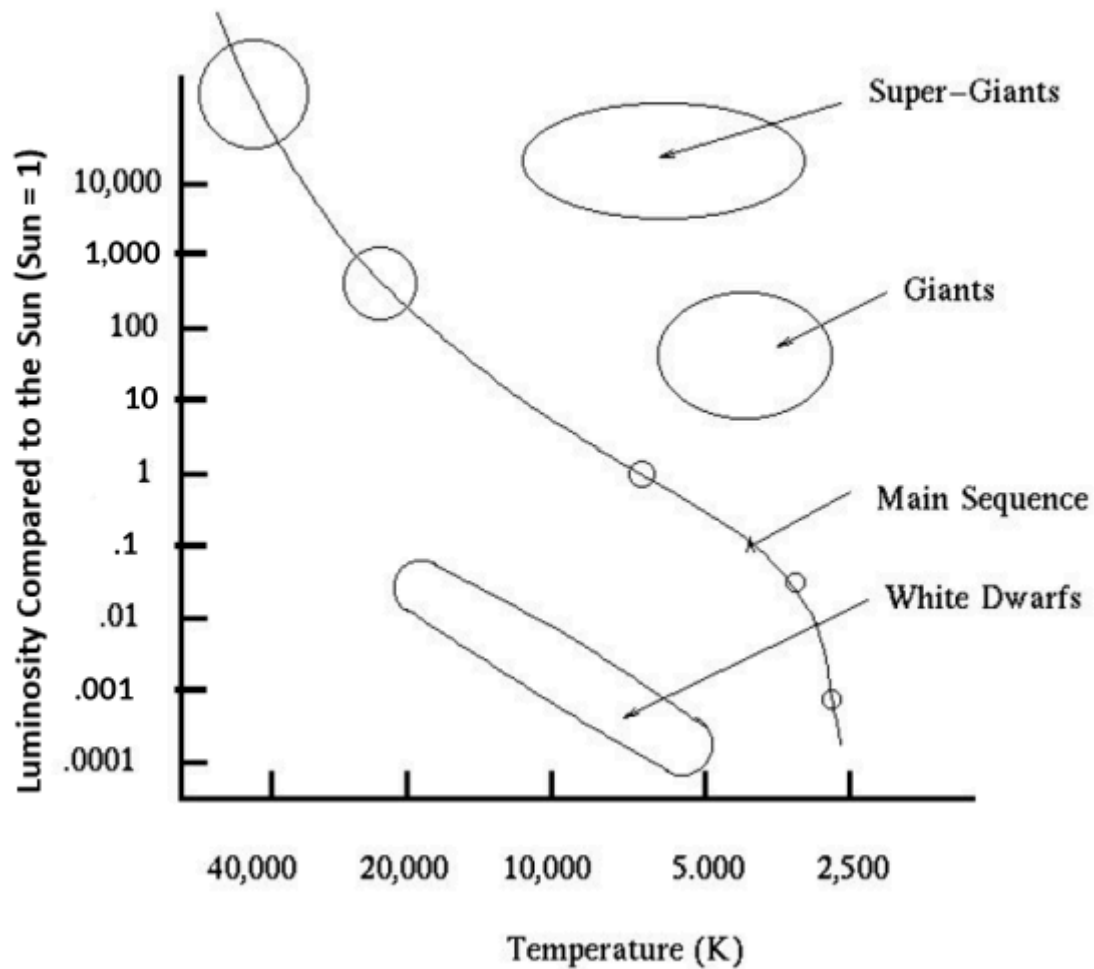
Table 1 below shows a comparison of the characteristics of our sun and two other stars in the Milky Way galaxy.

Table 1: Characteristics of Three Stars in the Milky Way Galaxy Table shows the comparison of three stars in the Milky Way Galaxy.			
	Our Sun	Betelgeuse	Proxima Centauri
Diameter	865,000 miles (1.4 million kilometers)	700 million miles (1.2 billion km)	130,488 miles (210,000 km)
Mass (In Solar Masses)	1	16.5 - 19	0.1221
Main color	Yellow	Orange-Red	Red
Luminosity (In number of Suns)	1	14,000	0.17
Surface Temperature	5,300 K - 6,000 K	3,500 K	3,000 K

Our sun, Betelgeuse, and Proxima Centauri were formed in a similar way. However, they will follow different paths in their life cycles.

Astronomers can determine where a star is in its life cycle by using certain characteristics of the stars and a model called the Hertzsprung-Russell Diagram.

Image 2: Hertzsprung-Russell Diagram



1 Which row correctly describes the life stage of our sun, Betelgeuse, and Proxima Centauri?

Row	Star	Life Stage
A	Sun	Giant
	Betelgeuse	Super Giant
	Proxima Centauri	White Dwarf
B	Sun	Main Sequence
	Betelgeuse	Super Giant
	Proxima Centauri	White Dwarf
C	Sun	Super Giant
	Betelgeuse	Main Sequence
	Proxima Centauri	Giant
D	Sun	Main Sequence
	Betelgeuse	Super Giant
	Proxima Centauri	Main Sequence

(1) Row A

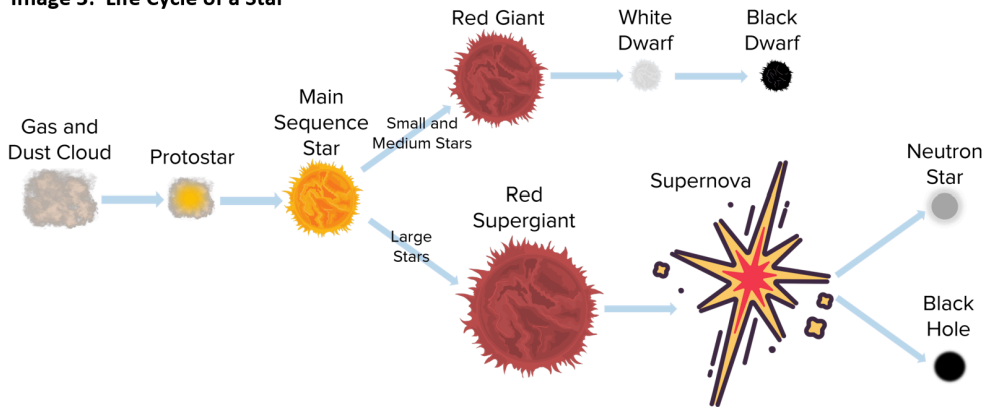
(2) Row B

(3) Row C

(4) Row D

Our sun, Betelgeuse, and Proxima Centauri follow a predicted path in the course of their lives. The diagram below represents possible pathways a star will follow.

Image 3: Life Cycle of a Star



2 A Examine the Life Cycle of a Star Model. Describe the similarities in the life cycles of our sun, Betelgeuse, and Proxima Centauri.

B Which of the three stars are on the same life cycle pathway? What caused the star(s) to diverge into different pathways?

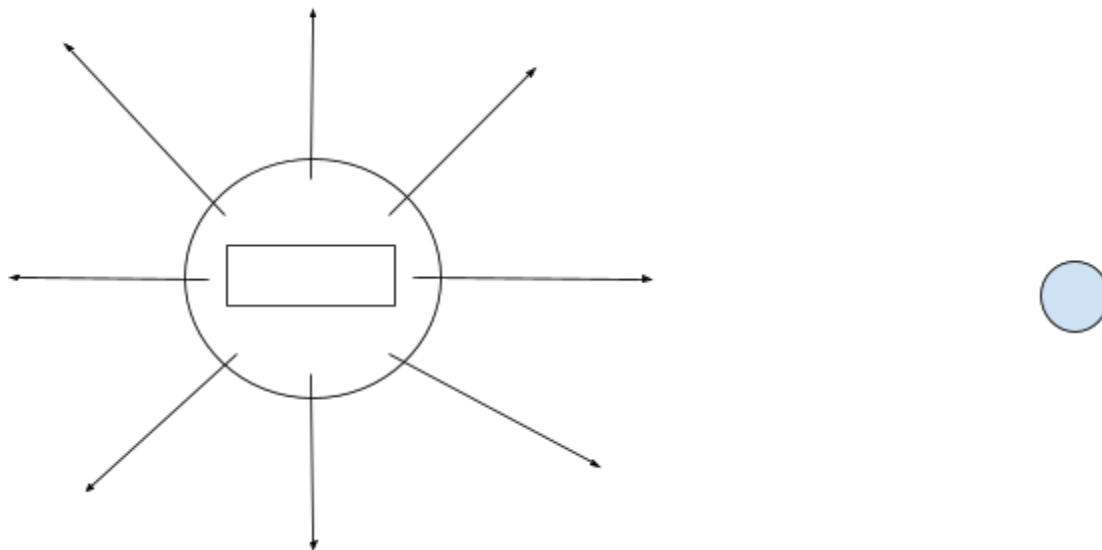
Betelgeuse, and Proxima Centauri.

Our sun and other stars undergo a process within their core that releases a tremendous amount of energy. This process has been able to release energy for 5 billion years and it is predicted that it will be able to release energy for another 5 billion years. Table 2 includes information about two possible energy releasing processes within our Sun's core.

Table 2: Possible Energy-Releasing Processes in the Sun		
Type of reaction	Equation	Estimated amount of time the Sun would be able to release energy
Chemical	$2\text{H}_2 + \text{O}_2 \longrightarrow 2\text{H}_2\text{O} + \text{Energy}$ 	50,000 years
Nuclear	$\text{Hydrogen} + \text{Hydrogen} \longrightarrow \text{Helium} + \text{Energy}$ 	10 billion years

3 Image 4 is an incomplete model of the Sun releasing energy that eventually reaches Earth in the form of radiation. Given that our Sun is predicted to release energy for a total of 10 billion years, use evidence from Table 2 to complete the model below by writing the process (chemical or nuclear) that releases energy from our Sun in the core. Be sure to cite your evidence in the space provided.

Image 4: Our Sun Releasing Radiation that Eventually Reaches Earth



Evidence: _____

The core temperatures of stars are much higher than their surface temperatures, due to the nuclear processes and element formations that take place there. Table 3 shows the core temperatures and elements formed for different life stages of stars.

Table 3: Element Production Through Nuclear Fusion in Stars					
	Yellow - Red Main Sequence	Red Giant	Red Supergiant	Massive Red Supergiant	Supernova
Core Temperature	$1.5 \times 10^7\text{K}$	$2 \times 10^8\text{K}$	$7 \times 10^8\text{K}$	$3 \times 10^9\text{K}$	$1 \times 10^{11}\text{K}$
Elements Formed	He	C, O, Ne, Mg	Na, Si, S, Ar, Ca	Fe, Ni	Elements with Atomic Number >28

4 A Based on the information in Table 3, what elements are currently formed in our sun, Betelgeuse, and Proxima Centauri?

Sun _____

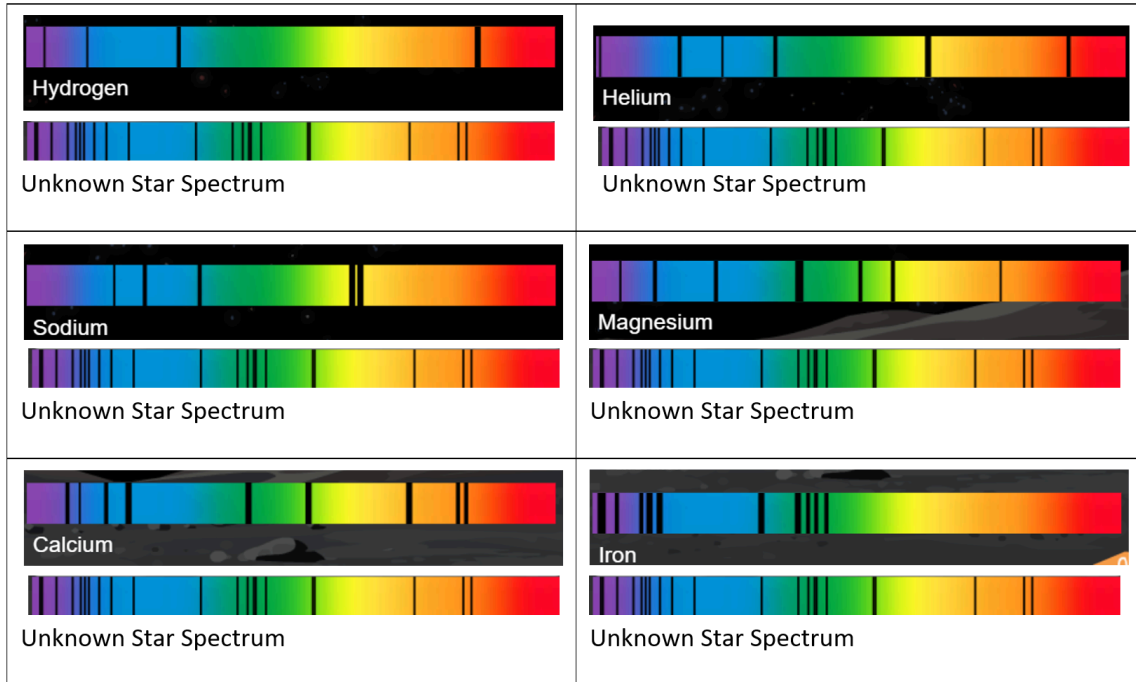
Betelgeuse _____

Proxima Centauri _____

B Which of the stars will produce elements heavier than iron in its lifetime? Provide reasoning for your claim. Use Image 3 and Table 3 to support your answer.

Spectrometers are tools that use a diffraction grating to split light into the different wavelengths of the visible light spectrum. Elements absorb specific wavelengths of light. The absorption of these wavelengths shows up as black lines on the spectrum. These lines can help determine what elements a star is composed of. Image 5 shows several element spectrum compared to the spectrum of an unknown star.

Image 5: Element Spectra Compared to an Unknown Star Spectrum



5 Which row in the table presents an accurate argument based on the evidence provided? Use Image 5 and Table 1 to answer the question.

Row	Claim	Evidence	Reasoning
A	The unknown star is Betelgeuse.	The spectra show that the star is composed of heavier elements like calcium and iron.	Betelgeuse is a red supergiant star. Red supergiant stars are massive enough to produce heavier elements.
B	The unknown star could be our sun or Betelgeuse.	The spectra show that the star is composed of sodium and magnesium.	Since both our sun and Betelgeuse both undergo nuclear fusion, both stars will produce the same elements at the same stage in their life cycle.
C	The unknown star is not our sun, Betelgeuse, or	The spectra show that the star is composed of iron.	Our sun and Proxima Centauri are in the early stages of their lives, so they

	Proxima Centauri.		have not reached the point where they are forming iron. Betelgeuse is a red supergiant star, which does not form anything heavier than calcium.
D	The unknown star could be our sun or Proxima Centauri.	The spectra show that the star is composed of hydrogen and helium.	Both our sun and Proxima Centauri are in the early stages of their life cycle, and have not started to form elements heavier than helium.

1. Row A
2. Row B
3. Row C
4. Row D

The nuclear fusion processes within our sun, Betelgeuse, and Proxima Centauri will eventually cease. The lifespan of stars is connected to the mass of the star. Table 4 shows the approximate lifespan of stars based on their masses.

Table 4: Star Masses vs. Lifespan	
Mass (1 Solar Mass = The Mass of Our Sun)	Time (Years)
0.1	1000s billion
1	10 billion
1.5	3 billion
3	370 million
10	32 million
30	11 million
60	3 million

6 Which set of terms constructs a correct explanation that describes the lifespan of our sun, Betelgeuse, and Proxima Centauri?

The more massive a star is, the _____ (A) _____ its lifespan is. Our sun's lifespan is approximately _____ (B) _____ years, Betelgeuse's lifespan is approximately _____ (C) _____ years, and Proxima Centauri's lifespan is approximately _____ (D) _____ years.

Row	A	B	C	D
1	longer	11 million	1000s billion	Between 3 billion and 10 billion
2	shorter	10 billion	Between 11 and 32 million	1000s billion
3	longer	32 million	370 million	3 million
4	shorter	Between 3 million and 11 million	60 million	10 billion

Rating Guide
A Star's Life Cluster

1 [1] Allow 1 credit for 4.

2 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

Part A: All three stars go through the first three stages, gas and dust cloud, protostar, and main sequence star.

Part B: Proxima Centauri and the sun will follow the same pathway. Stars start to follow a new pathway because of their masses. Betelgeuse is much more massive than the other two stars, so it will follow a different path than the sun and Proxima Centauri.

3 [1] Allow 1 credit for writing 'nuclear' in the core of the Sun, and for citing evidence of a chemical process only accounting for our Sun releasing energy for 50,000 years, while nuclear processes account for our Sun releasing energy for 10 billion years.

4 [1] Allow 1 credit for

Part A: Sun - He / Betelgeuse - Na, Si, S, Ar, Ca / Proxima Centauri - He

Part B: Acceptable responses include, but are not limited to:

Betelgeuse is the only star of the three that will produce elements heavier than iron. According to the life cycle of a star diagram more massive stars will explode in a supernova towards the end of their lives. Since Betelgeuse is currently a Red Supergiant, it will become a supernova. Table 1 shows that to produce elements heavier than iron, a star must explode in a supernova. Therefore, of the three stars, Betelgeuse is the only one that will produce elements heavier than iron in its lifetime.

5 [1] Allow 1 credit for 3.

6 [1] Allow 1 credit for 2.

Alignment Guide
A Star's Life Cluster

*Please note, any wording that is crossed out is not assessed in the item

Item	PE	DCI	CCC	SEP	PLD
1	HS-ESS1-1 Develop a model based on evidence to illustrate the life span of the sun and the role of nuclear fusion in the sun's core to release energy in the form of radiation.	ESS1.A-H1 The star called the Sun is changing and will burn out over a lifespan of approximately 10 billion years	Scale, Proportion, and Quantity	Developing and Using Models	NYS - Level 1: Given a model, and/or information, identify the evidence, from those provided, that determines the lifespan of the Sun or illustrates the role of nuclear fusion in the Sun's core to release energy or identify the correct pattern, from those provided, that shows variation in solar output.
Item	PE	DCI	CCC	SEP	PLD
2	HS-ESS1-1 Develop a model based on evidence to illustrate the life span of the sun and the role of nuclear fusion in the sun's core to release energy in the form of radiation.	ESS1.A-H1 The star called the Sun is changing and will burn out over a lifespan of approximately 10 billion years.	Stability and Change	Developing and Using Models	NYS - Level 2: Given a model, and/or information, describe the evidence that determines the lifespan of the Sun or illustrates the role of nuclear fusion in the Sun's core to release energy or ways that the Sun's radiation varies.
3	HS-ESS1-1 Develop a model based on	PS3.D-H1 Nuclear fusion processes	Matter and Energy;	Developing and Using	NYS - Level 3: Complete a model based on

	evidence to illustrate the life span of the sun and the role of nuclear fusion in the sun's core to release energy in the form of radiation.	in the center of the Sun release the energy that ultimately reaches Earth as radiation.	Scale, Proportion, and Quantity	Models	evidence to illustrate the lifespan of the Sun and/or the role of nuclear fusion in the Sun's core to release energy that eventually reaches Earth (in the form of radiation).
4	HS-ESS1-3 Communicate scientific ideas about the way stars, over their life cycle, produce elements.	ESS1.A-H4 Other than the hydrogen and helium formed at the time of the big bang, nuclear fusion within stars produces all atomic nuclei lighter than and including iron, and the process releases electromagnetic energy. Heavier elements are produced when certain massive stars achieve a supernova stage and explode.	Scale, Proportion, and Quantity	Obtaining, Evaluating, and Communicat ing Information	NYS - Level 5: Synthesize valid and reliable scientific information in order to communicate scientific ideas in multiple formats (i.e. orally, graphically, textually, or mathematically) about the way stars, over their life cycles, produce elements.
5	HS-ESS1-3 Communicate scientific ideas about the way stars, over their life cycle, produce elements.	ESS1.A-H2 The study of stars' light spectra and brightness is used to identify compositional elements of stars, their movements, and their distances from Earth.	Patterns	Analyzing and Interpreting Data; Obtaining, Evaluating, and Communicat ing Information	NYS - Level 5: Synthesize valid and reliable scientific information in order to communicate scientific ideas in multiple formats (i.e. orally, graphically, textually, or mathematically) about the way stars, over their life cycles, produce elements.
6	HS-ESS1-1 Develop a model based on evidence to illustrate the life span of the sun and	ESS1.A-H1: The star called the Sun is changing and will burn out over a lifespan of approximately 10 billion	Scale, Proportion, and Quantity	Constructing Explanations and Designing Solutions	NYS - Level 1: Given a model, and/or information, identify the evidence, from those provided, that

	the role of nuclear fusion in the sun's core to release energy in the form of radiation.	years.			determines the lifespan of the Sun or illustrates the role of nuclear fusion in the Sun's core to release energy or identify the correct pattern, from those provided, that shows variation in solar output.
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